

Function Prototypes

ELEC1006: ENGINEERING COMPUTING

Defining a Function

- A function is defined first before the main() function.
- Starting from the top, a compiler detects that `functionExample` is declared as a function and so the code below compiles successfully.

```
#include <iostream>
using namespace std;
void functionExample()
{
    cout << "In functionExample() function";
}
int main()
{
    functionExample();           // Call function
    return 0;                   // Exit program
}
```

Changing the Order

- Changing the order of the function below so that `main()` is declared before `functionExample()` yields an error in compilation.

```
#include <iostream>
using namespace std;
int main()
{
    functionExample();    // Call function
    return 0;             // Exit program
}
void functionExample()
{
    cout << "In functionExample() function";
}
```

Function Prototypes

- Ways to notify the compiler about a function before a call to the function:
 - Place function definition before calling function's definition
 - Use a function prototype (function declaration) – like the function definition without the body
 - **Prototype:** `void functionExample();`
 - **Header:** `void functionExample() { statements }`

Prototype Notes

- Place prototypes near top of program.
- Program must include either prototype or full function definition before any call to the function – otherwise there is compiler error.
- When using prototypes, can place function definitions in any order in source file.

Example

```
#include <iostream>
using namespace std;

void functionExample(); // Function prototype

int main()
{
    functionExample(); // Call function
    return 0;          // Exit program
}

void functionExample()
{
    cout << "In functionExample() function";
}
```

More info

- [1] cplusplus.com: Functions
<https://cplusplus.com/doc/tutorial/functions/>
- [2] learncpp.com: 2.7 – Forward declarations and definitions
<https://www.learncpp.com/cpp-tutorial/forward-declarations/>